Algebra Homework 4

Corrections to the problem sheet
Problem 2: replace $\mathbb{Q}[X, Y]$ with $F[X, Y]$ in the definition of $\mathfrak{a}$.
Problem 3: replace F^m with L^m in the definition of $V(S)$.

Problem 1. Let x be transcendental over $\mathbb{Q}$. Determine all $f \in \mathbb{Q}[X]$ such that $\mathbb{Q}(f(x)) = \mathbb{Q}(x)$.

Proof. Suppose that $\mathbb{Q}(f(x)) = \mathbb{Q}(x)$. Since $x \in \mathbb{Q}(f(x))$, then clearly f is nonconstant, and there are polynomials $P, Q \in \mathbb{Q}[X]$, with $\gcd(P, Q) = 1$ and $Q \neq 0$, such that

$$x = \frac{P(f(x))}{Q(f(x))}.$$

Since x is transcendental over $\mathbb{Q}$, this implies

$$XQ(f(X)) = P(f(X)).$$

Compute the degrees on both sides yields $1 + \deg Q \cdot \deg f = \deg P \cdot \deg f$. It follows that $1 = \deg f(\deg P - \deg Q)$. If $\deg f \geq 2$, the above equation is impossible. Together with that f is nonconstant, we have $\deg f = 1$. Conversely, for any polynomial with degree 1, write $f(X) = aX + b$ for some $a \in \mathbb{Q}^\times,\ b \in \mathbb{Q}$. Then $x = (f(x) - b)/a \in \mathbb{Q}(f(x))$, which completes the proof. □

Problem 2. Let F be a field in which -1 is not a square, and set

$$f = X^2 + Y^2, \qquad \mathfrak{a} = (f) \subset F[X, Y], \qquad B = F[X, Y]/\mathfrak{a}.$$

(i) Prove that f is irreducible.

(ii) Find a Noether basis of B over F.

Proof. (i) Suppose, for a contradiction, that f is reducible in $F[X, Y]$. Since f has total degree 2, we can write

$$X^2 + Y^2 = (aX + bY + c)(dX + eY + h),$$

for some $a, b, c, d, e, h \in F$.

Comparing constant terms gives $ch = 0$. If $c \neq 0$, then $h = 0$. Comparing the coefficients of X and Y now gives $cd = ce = 0$. Hence $d = e = 0$, contradiction. Thus $c = h = 0$. Therefore

$$X^2 + Y^2 = (aX + bY)(dX + eY).$$

Letting $Y = 1$ yields $X^2 + 1 = (aX + b)(dX + e)$. Comparing the coefficients of X^2 gives $ad = 1$, so $a \neq 0$. Consequently, $r = -b/a \in F$ is a root of $X^2 + 1$, and hence $r^2 = -1$. This contradicts the assumption that -1 is not a square in F. Thus f is irreducible in $F[X, Y]$.

Alternating Solution: Consider f as a primitive polynomial in X over the UFD $F[Y]$. By Gauss's lemma, it suffices to prove irreducibility over $F(Y)$. If f is reducible in $F(Y)$, it would have a root $r \in F(Y)$, so $r^2 + Y^2 = 0$. Then $(r/Y)^2 = -1$. Write $r/Y = u(Y)/v(Y)$ with $\gcd(u, v) = 1$ and $u, v \neq 0$. Then $u^2 = -v^2$. It follows that $\deg u = \deg v$ and comparing leading

coefficients gives $(u_0/v_0)^2 = -1$, where $u_0,\ v_0$ are the leading coefficients of $u,\ v$ respectively, which contradicts to the hypothesis. Hence f is irreducible in $F[X, Y]$.

(ii) Let $\overline{X}, \overline{Y}$ denote the class of X, Y in B. We prove that $\overline{X}$ is algebraically independent over F and that B is integral over $F[\overline{X}]$. Consider the natural homomorphism

$$\begin{aligned} \varphi : F[X] &\longrightarrow B, \\ g(X) &\longmapsto g(\overline{X}). \end{aligned}$$

We claim that φ is injective. Indeed, if $g \in \ker \varphi$, then $g(X) \in (X^2 + Y^2)$. Thus there exists $h \in F[X, Y]$ such that $g(X) = (X^2 + Y^2)h(X, Y)$. If $h \neq 0$, viewing both sides as polynomials in Y over $F[X]$ gives $\deg_Y((X^2 + Y^2)h) = 2 + \deg_Y h \geq 2$, whereas $\deg_Y g = 0$, a contradiction. Hence $h = 0$ and consequently $g = 0$. Thus φ is injective, so $\overline{X}$ is algebraically independent over F. Next, in B we have $\overline{Y}^2 = -\overline{X}^2$. Therefore $\overline{Y}$ satisfies the monic polynomial $T^2 + \overline{X}^2 \in F[\overline{X}][T]$, so it is integral over $F[\overline{X}]$. Hence $\{\overline{X}\}$ is a Noether basis of B over F. □

Problem 3. Let F be a field with algebraic closure L, and put $B_F = F[X_1, \ldots, X_m]$ and $B_L = L[X_1, \ldots, X_m]$. For $S \subseteq B_L$, define

$$V(S) = \{z \in L^m : g(z) = 0 \text{ for every } g \in S\}.$$

(i) If $\mathfrak{a} \subsetneq B_F$ is an ideal, prove that $V(\mathfrak{a})$ is nonempty and $\mathrm{Aut}(L/F)$-invariant.

(ii) If $W \subseteq L^m$ is $\mathrm{Aut}(L/F)$-invariant and $W = V(S)$ for some $S \subseteq B_L$, is $W = V(\mathfrak{a})$ for some ideal $\mathfrak{a} \subseteq B_F$?

(i) The nonemptiness of $V(\mathfrak{a})$ is the weak Hilbert Nullstellensatz; see Lang, *Introduction to Algebraic Geometry*, Chapter II, §4, Corollary to Theorem 7, p. 32. Its $\mathrm{Aut}(L/F)$-invariance follows immediately from the fact that the defining equations have coefficients in F.

(ii) Yes. This follows from the equivalence of conditions (C5) and (C7) in Lang, *Introduction to Algebraic Geometry*, Chapter III, §5, pp. 74–75.

Problem 4. let K/F be a field extension and $S \subseteq K$. Then S is algebraically independent over F if and only if

$$\text{for every } a \in S, \qquad F(S)/F(S \setminus \{a\}) \text{ is not algebraic.}$$

Proof. ($\Rightarrow$) Assume that S is algebraically independent over F, and fix $a \in S$. Put $E = F(S \setminus \{a\})$. Suppose, for a contradiction, that a is algebraic over E. Then there is a nonzero polynomial $f \in E[Y]$ with $f(a) = 0$. Its coefficients are rational functions of finitely many distinct elements $s_1, \ldots, s_n \in S \setminus \{a\}$. Clearing denominators gives a nonzero polynomial $P \in F[X_1, \ldots, X_n, Y]$ such that $P(s_1, \ldots, s_n, a) = 0$. This contradicts the algebraic independence of S. Thus a is transcendental over E, and $F(S) = E(a)$ is not algebraic over E.

($\Leftarrow$) Assume that $F(S)/F(S \setminus \{a\})$ is not algebraic for every $a \in S$. Suppose, for a contradiction, that S is algebraically dependent over F. Choose a finite dependent subset $T = \{s_1, \ldots, s_n\} \subseteq S$ of minimal cardinality, and put $T' = T \setminus \{s_n\}$. By minimality, T' is algebraically independent over F. Since T is dependent, there is a nonzero polynomial $P \in F[X_1, \ldots, X_n]$ with $P(s_1, \ldots, s_n) = 0$. Write

$$P(X_1, \ldots, X_n) = \sum_{j=0}^{d} P_j(X_1, \ldots, X_{n-1}) X_n^j.$$

At least one P_j is nonzero. By the independence of T', $P_j(s_1, \ldots, s_{n-1}) \neq 0$ whenever $P_j \neq 0$. Hence $g(Y) = P(s_1, \ldots, s_{n-1}, Y)$ is a nonzero polynomial in $F(T')[Y]$ with $g(s_n) = 0$. Thus s_n is algebraic over $F(T')$, and hence over $F(S \setminus \{s_n\})$. It follows that $F(S) = F(S \setminus \{s_n\})(s_n)$ is algebraic over $F(S \setminus \{s_n\})$, contradicting the hypothesis. Therefore S is algebraically independent over F. $\square$